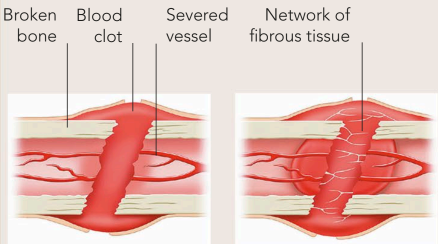
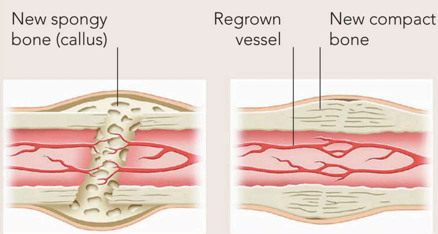
BONE REPAIR

Despite its image as dry, brittle, and even lifeless, bone is an active tissue with an extensive blood supply and its own restorative processes. After a fracture, blood clots as it does elsewhere in the body. Fibrous tissue, and then new bone growth, bridge the break and eventually restore strength. However, medical treatment is often required to ensure that the repair process is effective and the result is not misshapen. If the bones are displaced, manipulation to restore their normal position – known as reduction – may be performed under anaesthesia. The bone will also be immobilized to allow the ends to heal correctly.



IMMEDIATE RESPONSE
Blood leaks from the blood vessels and clots. White blood cells gather at the area to scavenge damaged cells and debris.

AFTER SEVERAL DAYS
Fibroblast cells construct new fibrous tissue across the break. The limb is immobilized, usually in a plaster cast or splint.

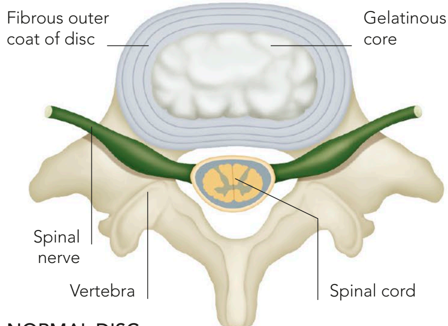


AFTER 1–2 WEEKS
Bone-building cells (osteoblasts) multiply and form new bone tissue. Initially spongy, the new tissue infiltrates the site of the fracture as a callus.

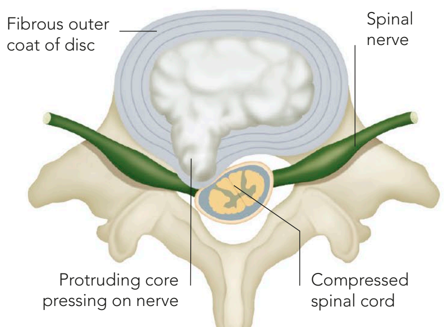
AFTER 2–3 MONTHS
Blood vessels reconnect across the fracture. The callus reshapes while the new bone tissue is "remodelled" into dense, compact bone.

DISC PROLAPSE

The cushion-like cartilage discs that separate adjacent vertebrae have a hard outer covering and a jelly-like centre. An accident, wear and tear, or excessive pressure when lifting awkwardly, may rupture the outer layer. This forces some of the core material to bulge out, or prolapse. The prolapsed (or herniated) portion may cause pressure on the nearby spinal nerve root. Symptoms of disc prolapse include dull pain, muscle spasm and stiffness in the affected area of the back, and pain, tingling, numbness, or weakness in the body part supplied by the nerve.



NORMAL DISC
The outer casing (capsule) of the intervertebral disc is intact and encloses its gelatinous core. The disc sits between the bodies (centra) of adjacent vertebrae.



PROLAPSED DISC
A weak site in the outer casing allows the gelatinous core to bulge through as the disc is compressed. The resulting pressure on the spinal nerve causes pain.